

Sea $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ una transformación lineal tal que su matriz en la base canónica E es:

$$M_{EE}(T) = \begin{pmatrix} 0 & k & 1 \\ k & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

a) Hallar $k \in \mathbb{R}$, si existe, tal que $\dim(N(T)) = 1$.

b) Para $k = 2$, hallar $R(T)$.

$$\text{rang}(M_{EE}(T))$$

a) T. Dimension: $\dim(V) = \dim(N(T)) + \dim(R(T))$

$$V = \mathbb{R}^3 \Rightarrow 3 = 1 + \text{rang}(M_{EE}(T)) \Rightarrow \underline{\text{rang}(M_{EE}(T)) = 2}$$

$$\det(M_{EE}(T)) = \begin{vmatrix} 0 & k & 1 \\ k & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = 0$$

$$k + k = 0 \Rightarrow 2k = 0 \Rightarrow \underline{k = 0} \Rightarrow M = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

b) Si $k = 2$

$$M_{EE}(T) = \begin{pmatrix} 0 & 2 & 1 \\ 2 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \Rightarrow \det(M_{EE}(T)) = 2k = 4 \neq 0 \text{ tiene } \underline{\text{rang} = 3}$$

$$R(T) = \mathbb{R}^3$$

Sea $A = \begin{pmatrix} 3 & 0 & 1 \\ 1 & k & -1 \\ 2 & 0 & 2 \end{pmatrix}$.

a) Hallar $k \in \mathbb{R}$ para que $\lambda = -1$ sea autovalor de A .

b) Para $k = 2$, diagonalice la matriz.

a) $\det(A - \lambda I) = 0$

$$\begin{vmatrix} 3 - (-1) & 0 & 1 \\ 1 & k - (-1) & -1 \\ 2 & 0 & 2 - (-1) \end{vmatrix} = \begin{vmatrix} 4 & 0 & 1 \\ 1 & k+1 & -1 \\ 2 & 0 & 3 \end{vmatrix} = (k+1) \cdot (-1)^{2+2} \begin{vmatrix} 4 & 1 \\ 2 & 3 \end{vmatrix} = 0$$

$$(k+1) \cdot 10 = 0 \Rightarrow \underline{k = -1}$$

b) $k=2$

$$A = \begin{pmatrix} 3 & 0 & 1 \\ 1 & 2 & -1 \\ 2 & 0 & 2 \end{pmatrix} \Rightarrow \boxed{A = P \cdot D \cdot P^{-1}}$$

$$\det(A - \lambda I) = \begin{vmatrix} 3-\lambda & 0 & 1 \\ 1 & 2-\lambda & -1 \\ 2 & 0 & 2-\lambda \end{vmatrix} = (2-\lambda) \cdot \begin{vmatrix} 3-\lambda & 1 \\ 2 & 2-\lambda \end{vmatrix} = 0$$

$$(2-\lambda) \cdot [(3-\lambda) \cdot (2-\lambda) - 2] = 0$$

$$(2-\lambda) \cdot [6 - 3\lambda - 2\lambda + \lambda^2 - 2] = 0$$

$$(2-\lambda) [\lambda^2 - 5\lambda + 4] = 0$$

$$(2-\lambda) \cdot (\lambda-1) \cdot (\lambda-4) = 0$$

$$\lambda_1 = 2, \quad \lambda_2 = 1, \quad \lambda_3 = 4$$

$$M_A = 1, \quad M_A = 1, \quad M_A = 1$$

⊗ Autovectores asociados: $(A - \lambda I) \vec{v} = 0$

$$\lambda_1 = 2 \quad \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & -1 \\ 2 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} x + z = 0 \\ x = 0 \end{cases}$$

$$\vec{v}_1 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad M_B = 1$$

$$\lambda_2 = 1 \quad \begin{pmatrix} 2 & 0 & 1 \\ 1 & 1 & -1 \\ 2 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} 2x + z = 0 \\ x + y - z = 0 \end{cases}$$

$$\vec{v}_2 = \begin{pmatrix} 1 \\ -3 \\ -2 \end{pmatrix} \Rightarrow M_B = 1$$

$$\lambda_3 = 4 \quad \begin{pmatrix} -1 & 0 & 1 \\ 1 & -2 & -1 \\ 2 & 0 & -2 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} -x + z = 0 \\ x - 2y - z = 0 \end{cases}$$

$$\vec{v}_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \Rightarrow M_B = 1 \quad \text{Como } M_A = M_B. \text{ La matriz es DIAGONALIZABLE}$$

$$D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix} ; \quad P = \begin{pmatrix} 0 & 1 & 1 \\ 1 & -3 & 0 \\ 0 & -2 & 1 \end{pmatrix}$$

Dada la matriz $A = \begin{pmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}$.

d) Factoriz. SVD: $A = U \cdot \Sigma \cdot V^T$; $U \in \mathbb{R}^{3 \times 2}$; $\Sigma \in \mathbb{R}^{2 \times 2}$, $V^T \in \mathbb{R}^{2 \times 2}$

$$A^T \cdot A = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

$2 \times 2 \qquad \qquad 3 \times 2$

• Autovalores: $\det(A^T A - \lambda I) = 0$

$$\begin{vmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{vmatrix} = (2-\lambda)^2 - 1 = 0$$

$$\begin{cases} |2-\lambda| = 1 \\ 2-\lambda = 1 \\ 2-\lambda = -1 \end{cases}$$

$$\frac{\lambda_1=3}{\sigma_1=\sqrt{3}} , \frac{\lambda_2=1}{\sigma_2=1} \Rightarrow \Sigma = \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \end{pmatrix}$$

• Autovectores (matriz V)

$\lambda=3 \quad (A^T A - 3I) \cdot \vec{v} = \vec{0}$

$$\begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{matrix} -x - y = 0 \\ y = -x \end{matrix} \Rightarrow \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\|\vec{v}\| = \sqrt{2} \Rightarrow \vec{v}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$\lambda_2 = 1 \quad (A^T A - 1I) \cdot \vec{v} = \vec{0}$

$$\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow x - y = 0 \Rightarrow x = y \Rightarrow \vec{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\|\vec{v}\| = \sqrt{2} \Rightarrow \vec{v}_2 = \frac{1}{\sqrt{2}} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} \Rightarrow V = \frac{1}{\sqrt{2}} \cdot \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

* Hallamos U : $u_i = \frac{1}{\sigma_i} \cdot A \cdot v_i$

$$u_1: u_1 = \frac{1}{\sqrt{3}} \cdot \begin{pmatrix} 1 & -1 \\ 0 & 0 \\ 1 & 1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

$$u_2 = \frac{1}{1} \cdot \begin{pmatrix} 1 & -1 \\ 0 & 0 \\ 1 & 1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$$U = \begin{pmatrix} 2/\sqrt{6} & 0 \\ -1/\sqrt{6} & 1/\sqrt{2} \\ 1/\sqrt{6} & 1/\sqrt{2} \end{pmatrix} \quad A = U \cdot \Sigma \cdot V^T$$

Forma LARGA. $\Rightarrow U \in \mathbb{R}^{3 \times 3}$; $\Sigma \in \mathbb{R}^{3 \times 2}$ $V \in \mathbb{R}^{2 \times 2}$

Para hallar u_3 hacemos el prod. vectorial entre u_1 y u_2 :

$$u_3 = \frac{u_1 \times u_2}{\|u_1 \times u_2\|} = \frac{1}{\sqrt{3}} \cdot \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$$

$$\begin{vmatrix} i & j & k \\ 2/\sqrt{6} & -1/\sqrt{6} & 1/\sqrt{6} \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{vmatrix}$$

$$U = \begin{pmatrix} 2/\sqrt{6} & 0 & -1/\sqrt{3} \\ -1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \end{pmatrix}; \quad \Sigma = \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}; \quad V^T = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

b) Factoriz. QR ($A = Q \cdot R$) Dada la matriz $A = \begin{pmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}$.

$$a_1 = (1, 0, 1)^T \quad a_2 = (-1, 1, 0)^T$$

• Hallar Q :

$$\|a_1\| = \sqrt{2} \Rightarrow q_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

• Proyectamos a_2 sobre q_1 y restamos

$$\text{Proy}_{q_1}(a_2) = (q_1^T \cdot a_2) q_1 \Rightarrow q_1^T \cdot a_2 = -\frac{1}{\sqrt{2}}$$

$$p_2 = a_2 - \text{proy}_{f_1}(a_2) = \begin{pmatrix} -0,5 \\ 1 \\ 0,5 \end{pmatrix} \Rightarrow \|p_2\| = \sqrt{\frac{3}{2}}$$

$$q_2 = \frac{p_2}{\|p_2\|} = \frac{1}{\sqrt{6}} \cdot \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$$

q_3 tiene que ser ortogonal a f_1 y f_2 . Hacemos el prod. vect.

$$q_3 = f_1 \times f_2 \Rightarrow q_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

$$Q = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & -1/\sqrt{3} \\ 1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \end{pmatrix} ; \quad R = Q^T \cdot A = \begin{pmatrix} \sqrt{2} & -\frac{1}{\sqrt{2}} \\ 0 & \sqrt{3/2} \\ 0 & 0 \end{pmatrix}$$

NOTA: $r_{11} = \|a_1\| = \sqrt{2}$; $r_{12} = q_1^T \cdot a_2 = -\frac{1}{\sqrt{2}}$; $r_{22} = \|p_2\| = \sqrt{\frac{3}{2}}$

